

ARJUN VARMA

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EDUCATION

Columbia University

Master of Science in Data Science

Coursework: Applied Machine Learning, Agentic AI for Analytics, Statistical Inference and Modeling, Probability and Statistics
Teaching Assistant, Columbia Business School: Business Analytics II (Foundations of AI) and Hollywood and Big Data

New York, NY

Dec 2026

Vellore Institute of Technology

Bachelor of Technology in Electronics and Communication Engineering

Special Achiever Award | Merit Scholarship

Vellore, IN

May 2022

WORK EXPERIENCE

Novo Nordisk

Data Scientist Intern

- Improving next-best-engagement models and supporting pre-launch analytics for denecimig, informing HCP targeting and field readiness
- Building an LLM layer over unstructured HCP and field inputs that surfaces key engagement drivers across access, efficacy, competitor prescribing, and GLP-1 adoption, and translates them into tailored field guidance

Plainsboro, NJ

Jun 2026 - Aug 2026

ZS Associates

Advanced Data Science Associate Consultant

- Built and deployed an org-wide analytics and ML platform (Spark/SQL, dashboards) that unified 5+ data sources into territory and product KPIs used by 100+ stakeholders supporting a \$10B oncology portfolio
- Developed a biliary tract cancer early-detection model across 250M patient claims using clinical risk factors, K-means/GMM segmentation, and Transformer-based NLP clustering, enabling ~45-day earlier identification vs standard diagnosis lag; presented at PMSA 2025 and adopted for territory-level resource planning
- Cut weekly reporting time from days to minutes, replacing Excel workflows with automated pipelines and self-serve dashboards

Pune, IN

Feb 2025 - Jun 2025

ZS Associates

Decision Analytics Associate Consultant

- Built and deployed Positive-Unlabeled learning models to infer missing categorical labels in medical transaction data, increasing customer-journey analytics coverage from ~40% to ~95% across tumor types and territories
- Led a 5-member team to modernize legacy business rules; saved ~50 hrs/mo and improved first-pass quality to >99%
- Placed in the top ~10% in a company-wide hackathon and earned selection for a lateral transfer into the Data Science vertical

Pune, IN

Jul 2024 - Jan 2025

ZS Associates

Decision Analytics Associate

- Delivered ad hoc analyses identifying care gaps and market opportunities to inform brand strategy across multiple launches
- Engineered PySpark/SQL ETL pipelines integrating IQVIA and Symphony claims, Flatiron EHR, and Texas Oncology practice data covering millions of patients for \$4B+ oncology drug performance analytics
- Defined patient cohort inclusion and exclusion logic robust to missing and miscoded fields, enabling audit-ready reporting
- Promoted to Associate Consultant in 4 cycles (typical: 5) and received Expert Associate and Insight Illuminator awards

Pune, IN

Feb 2022 - Jun 2024

PROJECT EXPERIENCE

Airbnb Data Analyst Agent - Python, LangChain, FastAPI, DuckDB, Next.js

- Built a multi-agent data analyst (Planner, SQL, Validator, Chart, Narrator agents on a typed message bus) turning natural-language questions into validated SQL, charts, and narrated answers with row-level citations
- Engineered guardrails (SQL dry-runs, row-count and type audits, self-critique with retries) plus a golden Q/A regression eval suite and per-query latency and cost tracking

[GitHub](#) / [Live Demo](#)

Financial RAG Chatbot - Python, FastAPI, ChromaDB, Streamlit, GCP

- Built an LLM-powered RAG chatbot answering financial questions from unstructured SEC filings with line-level citations, using ChromaDB semantic retrieval, text-embedding-3-large, and automatic ticker and period parsing
- Built a multi-model evaluation pipeline using Claude Opus as a judge and deployed a live demo on Streamlit Cloud

[GitHub](#)

Agricultural Product Standardization & Risk Detection - SunCulture (Series B Agtech) - Python, RAG, REST API

- Built a RAG-augmented classification system categorizing 7M+ farmer transactions across 500+ product categories to support creditworthiness assessment for microloans in East Africa
- Achieved 99% accuracy on a 10K-item holdout via hybrid rule-based + LLM-assisted classification, cutting manual review by 95%

DEUCE: Tennis Forecast Engine (ATP + WTA) - Python, XGBoost, Next.js, GitHub Actions

- Built a hybrid forecast engine over 88K+ tour matches combining surface Elo with cross-surface transfer, an opponent-adjusted serve/return Markov point model, and a Platt-calibrated XGBoost combiner with Monte Carlo draw simulation
- Beat the bookmaker benchmark on walk-forward, leakage-free ATP predictions (0.195 vs 0.196 Brier, 69.6% vs ~69% accuracy); ships with hourly data refresh, daily retrains, and 182 CI tests via GitHub Actions

[GitHub](#) / [Live Site](#)

SKILLS

Programming: Python, SQL, R, C++

Analytics & ML: classification, regression, NLP and text analytics, clustering and segmentation, LLMs/RAG, prompt engineering

Data & MLOps: Spark (PySpark), Databricks, ETL/ELT, MLflow, Docker, AWS (S3, EMR, Athena, SageMaker)

Libraries: pandas, NumPy, scikit-learn, PyTorch, SHAP, matplotlib, BeautifulSoup

Reporting: Excel (advanced), PowerPoint, stakeholder reporting, self-serve dashboards

Tools: Git, Linux, CI/CD, Jupyter, Jira, Confluence, Claude Code, Cursor IDE